

Pavan Kumar Adapala

DevOps Engineer

pavankumar.adapala.msc@gmail.com | +49 15751763888 | 70435 Stuttgart, Germany | [LinkedIn](#) | [Portfolio](#)

Profile Summary

DevOps Engineer with around **3 years of experience** in provisioning scalable **on-premises** and **cloud infrastructures**, developing **GitOps-driven CI/CD pipelines** for **containerized applications**, and implementing observability solutions to enhance deployment reliability and reduce operational costs. Proven success in reducing **debugging time** by up to **30%**, enabling faster and more stable application delivery. Skilled in **AWS, Azure, Jenkins, Terraform, Docker, Kubernetes, Grafana, Bash** and **Python** with a strong focus on automation, security, and observability.

Professional Experience

Data Analyst – Masters Thesis | *Dürr Systems AG, Bietigheim-Bissingen, Germany* 02/2024 - 07/2024

- Developed and evaluated **ML-based load forecasting models** (batch and one-step-ahead) using **IIoT time-series data** from automotive paint shop HVAC systems. These models enhanced **Predictive Energy Management (PEM)** by achieving **95% forecasting accuracy**, accounting for weather-related uncertainties, and enabling proactive load shifting to **reduce energy waste and operational costs**.

Software Integrator – Research Internship | *Robert Bosch GmbH, Stuttgart, Germany* 05/2023 - 01/2024

- Designed and **optimized Jenkins CI/CD pipelines** to enhance the development process of **autonomous parking** software, including support for build and integration issues.
- Implemented **data-driven automation strategies** using **Python** and **REST APIs** to parse CI/CD **build artifacts** and automatically email key test summaries (unit, smoke, integration), reducing debugging time by **30%** and accelerating root cause identification.
- Built a **centralized Grafana dashboard** leveraging **PostgreSQL** queries to monitor CI/CD **build KPIs** across multiple projects, enabling **data-driven release planning**.

Trainee DevOps Engineer | *Ibexlabs Cloud Consulting Pvt Ltd, Hyderabad, India* 04/2021 - 09/2021

- Provisioned and managed AWS infrastructure using **Terraform** and CloudFormation to support **development** and **operations teams**, ensuring scalable, repeatable deployments.
- Developed custom **CloudWatch** dashboards and **alerting systems** by **integrating open-source monitoring agents**, enabling proactive application management, and reducing **downtime**.

Cloud Engineer | *Isuzu Motors India Pvt Ltd, Sri City, India* 06/2018 - 07/2019

- Contributed to **cloud migration** proof-of-concept (PoC) project, **reducing operational cost** by **25%** through implementation of scalable (Auto Scaling), production-ready cloud infrastructure.
- Implemented **CI/CD** pipelines using **Jenkins Master-Slave architecture** to automate deployments, **distributing workloads** for faster builds and accelerating release cycles.
- Deployed the **web application** following **3-tier AWS architecture**, ensuring secure, highly available (HA), and fault-tolerant systems.

Education

Master of Science in Scientific Instrumentation (M.Sc.) 10/2021 - 06/2025
University of Applied Sciences Jena, Germany **GPA 1,6**

Bachelor of Technology in Mechanical Engineering (B.Tech.) 09/2014 - 04/2018
Gudlavalleru Engineering College, India **GPA 1,7**

Skills

Programming, APIs: Python, Bash scripting, React, TypeScript | FastAPI

Cloud & Infrastructure: AWS (IaaS, PaaS, SaaS), Azure | Terraform, CloudFormation

CI/CD & GitOps Automation: Jenkins, GitHub Actions | ArgoCD | Git, GitHub, Bitbucket | Jira, Confluence

Containers & Platform Engineering: Docker, Podman, Kubernetes | Ansible, Helm

Virtualization & Operating systems: VMware, VirtualBox | Linux administration (Ubuntu, RHEL 9)

Observability & Reliability: Prometheus, Grafana, AWS CloudWatch

Databases & Web Servers: PostgreSQL, MySQL, Amazon (RDS, DynamoDB) | Nginx, Apache

Security & Networking: IAM, Trivy, SonarQube | TCP/IP, DNS, HTTP/HTTPS | VPC, Route53, Load balancer

Projects

End-to-End MLOps Platform for Energy Consumption Prediction, [GitHub Link](#) 12/2025

- Designed and operated **production-grade platforms** spanning cloud VMs, containerized services, and Kubernetes environments, with a strong focus on automation, and operational reliability.
- Developed **MLOps pipeline**, covering data ingestion, model development, containerized inference services, and cloud deployment, enabling ML models to be reliably operated in production.
- **Developed and integrated RESTful APIs** (FastAPI) for ML inference and backend services, ensuring clean frontend (**React**)–backend separation and scalable, **service-oriented application architecture**.

Centralized Monitoring & Performance Optimization for Hybrid Infrastructure, [GitHub Link](#) 06/2025

- Built centralized **Prometheus–Grafana monitoring** for a hybrid infrastructure (cloud VMs, Kubernetes clusters, on-prem RHEL servers), proactively **detecting CPU throttling and memory bottlenecks** at pod and VM level, enabling **preemptive resource adjustments** that prevented production outages and pod restarts.

End-to-End GitOps-Driven CI/CD Pipeline for Web Application Deployment, [Project Link](#) 05/2025

- Containerized the application with a **multi-stage Dockerfile** and developed GitOps-driven CI/CD pipeline by integrating **Argo CD** into Kubernetes cluster to enable declarative, automated deployments, and improving reliability.
- Implemented **DevSecOps** practices using **SonarQube** (SAST), and **Trivy** (container vulnerability scanning), ensuring secure and high-quality application releases.

Language and Soft Skills

- English (fluent) - C1, German (conversational) - B1, Telugu (native) - C2
- Team collaboration, problem-solving, creativity and innovation, quick learning

Certifications

- Red Hat Certified System Administrator (RHCSA) - Pluralsight 12/2025
- AWS Certified Developer Associate 03/2023
- AWS Certified Cloud Practitioner 03/2022